

FEDCARE: Client-Adaptive Curriculum and Reference Evolution for Federated Direct Preference Optimization

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Abstract

Direct Preference Optimization (DPO) aligns language models with pairwise preferences, but its federated application must accommodate heterogeneous client distributions, unequal usefulness of preference pairs, and changing personalized policies. We propose FEDCARE, a federated DPO framework that couples client-adaptive curriculum construction with personalized reference refresh. Each client ranks eligible preference pairs using a policy-dependent margin: an initial-policy preference gap in the first round and a change in that gap between successive personalized checkpoints in later rounds. Under a Bradley-Terry clean-preference model and bounded implicit-reward approximation error, we derive a margin-dependent lower bound on preference validity and an upper bound on the associated gradient discrepancy. After scoring and selection, each client refreshes its reference from the current pre-update personalized checkpoint and freezes it throughout local DPO. We adopt the selective LoRA-sharing mechanism of FedSA-LoRA to aggregate LoRA-A while retaining LoRA-B locally. Under the reported evaluation protocol, FEDCARE achieves 70.26% macro-average accuracy on MATH and 24.24 macro-average BLEU on CodeXGLUE, exceeding the strongest compared baselines by 1.90 percentage points and 3.47 BLEU points, respectively. Additional results under cross-domain heterogeneity, varying client scales, and multiple backbone models further demonstrate the effectiveness and robustness of FEDCARE.

CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

federated learning, direct preference optimization, large language models, curriculum learning

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1 Introduction

Large language models (LLMs) have shown strong capabilities in reasoning-intensive tasks such as mathematical problem solving, code generation, and logical inference [6, 36]. Unlike general instruction following, these tasks require models to maintain coherent intermediate steps, long-range consistency, and reliable solution verification [22, 35]. Strengthening the reasoning ability of LLMs has thus become a central goal of post-training and alignment.

Preference optimization offers an effective route to this goal. Rather than merely imitating reference answers, it trains the model to assign higher likelihood to preferred responses than to rejected

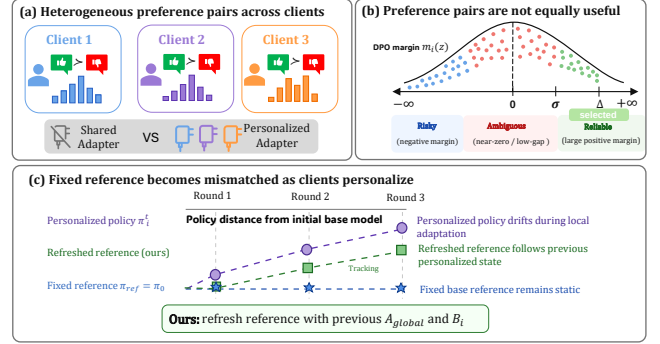


Figure 1: Motivation of the proposed method. Federated DPO for LLMs faces three coupled challenges: heterogeneous client preference distributions, unequal usefulness of preference pairs, and stale reference policies across communication rounds.

ones [5, 28, 32]. Direct Preference Optimization (DPO) streamlines this idea by optimizing the policy directly on pairwise preference data, removing the need for an explicit reward model or reinforcement learning [31].

Most preference optimization methods, however, assume that all preference data are centrally available [3, 39]. This assumption rarely holds for reasoning-oriented applications, where data are split across clients, institutions, or domains under privacy and ownership constraints [18, 20]. Federated learning is a natural remedy, but applying DPO directly in a federated setting raises three coupled challenges, illustrated in Fig. 1. First, preference distributions are heterogeneous: clients differ in reasoning domains, difficulty levels, and preference criteria, so a single shared adapter cannot serve them all [33]. Second, preference pairs are not equally useful: pairs with large model-aware margins provide more reliable preference signals, whereas low-margin or ambiguous pairs are more likely to contain uncertain, noisy, or unstable supervision [8]. Third, a fixed DPO reference can become stale as the personalized client policy changes across communication rounds, leaving subsequent local updates anchored to an earlier policy state [24]. This motivates a client-specific reference refresh schedule that preserves the temporal comparison needed for curriculum scoring while updating the reference before local optimization.

To address these challenges, we propose FEDCARE, a federated DPO framework based on Client-Adaptive Curriculum and Reference Evolution for LLMs. In contrast to conventional federated DPO, which trains on a fixed local preference set, FEDCARE treats curriculum construction as a dynamic, client-specific process: each client scores candidate preference pairs by their usefulness and reliability

using its current policy, reference model, and DPO-margin distribution. As training proceeds, the selected curriculum expands from reliable samples toward more challenging ones, while respecting local heterogeneity and privacy.

Our contributions are summarized as follows:

- We propose FEDCARE, a federated direct preference optimization framework that integrates FedSA-LoRA-style personalization with client-adaptive curriculum selection and an update-after-scoring personalized reference refresh schedule.
- We design a *Client-Adaptive Curriculum Construction* strategy that ranks eligible preference pairs using client-specific DPO margins and progressively introduces challenging samples. We couple this strategy with *Dynamic Personalized Reference Evolution (DPRE)*, an update-after-scoring personalized reference refresh schedule, to achieve personalized and stable federated DPO.
- We empirically validate FEDCARE on heterogeneous MATH and CodeXGLUE benchmarks, demonstrating consistent improvements over representative federated preference optimization baselines. Further analyses verify that the gains come from adaptive preference selection, personalized parameter sharing, and evolving reference policies, rather than stronger aggregation alone.

2 Related Work

2.1 Federated Parameter-Efficient Adaptation

Federated learning (FL) lets clients train collaboratively without sharing private data. FedAvg aggregates local updates on the server [26], but suffers from client drift under non-IID data. FedProx introduces proximal regularization to stabilize local optimization under heterogeneous data [21], and recent surveys summarize heterogeneity, communication, privacy, and personalization as central FL challenges [18]. Personalized FL mitigates client drift by balancing global sharing against local specialization [33]; representative approaches include FedPer and FedRep, which separate shared representations from client-specific components [1, 7]. Because full-parameter federated fine-tuning is prohibitive for LLMs, a line of work instead updates LoRA adapters over a frozen backbone [14, 25, 40]. Most relevant to us, FedSA-LoRA observes that LoRA-A and LoRA-B play asymmetric roles and shares only LoRA-A across clients [11]. FEDCARE adopts this selective-sharing principle but, unlike prior federated LoRA work aimed at instruction tuning, integrates it with preference optimization and margin-driven curriculum construction for LLMs.

2.2 Federated Preference Optimization

Preference optimization aligns LLMs with human or task-specific preferences by favoring preferred responses over rejected ones. RLHF first trains reward models from human comparisons and then optimizes policies with reinforcement learning [5, 28, 32]. DPO removes the explicit reward model by optimizing pairwise preferences through a reference-regularized objective [31]. Subsequent methods further analyze or simplify preference optimization, such as IPO, KTO, and SimPO [2, 9, 27]. Recent studies begin to connect alignment with federated learning. OpenFedLLM provides a

benchmark and framework for decentralized LLM fine-tuning [39]. FEDCARE is complementary: rather than fixing the local preference set, each client dynamically selects useful pairs based on its own policy and reference model.

2.3 Curriculum Preference Data Selection

Curriculum learning improves training by ordering samples meaningfully, typically from easy to hard [4]. Related ideas such as self-paced learning and competence-based curricula further show that adaptive data schedules can improve optimization stability [19, 30]. In preference optimization, difficulty-aware methods construct or select preference pairs according to their learning signal. Curry-DPO introduces ranked preferences through curriculum learning [29]. Less is More similarly shows that selecting high-quality preference data can improve alignment efficiency [8]. In federated settings, FibecFed explores curriculum learning for LLM fine-tuning under client heterogeneity [23]. Yet these methods either assume centralized preference data or do not address DPO-style preference optimization, which motivates our client-specific margin-driven curriculum.

3 Method

3.1 Problem Formulation

We consider a federated preference optimization setting with M heterogeneous clients. Client i owns a local preference dataset

$$\mathcal{P}_i = \{(x_{i,j}, y_{i,j}^w, y_{i,j}^l)\}_{j=1}^{n_i}, \quad (1)$$

where $x_{i,j}$ is an input prompt, $y_{i,j}^w$ is the preferred response, and $y_{i,j}^l$ is the rejected response.

Let W_0 denote a frozen pretrained backbone. For each adapted layer ℓ , client i uses the Dual-LoRA parameterization

$$W_{i,\ell} = W_{0,\ell} + B_{i,\ell}A_\ell, \quad (2)$$

where A_ℓ is shared across clients and $B_{i,\ell}$ remains client-specific. We denote their collections by $\mathcal{A} = \{A_\ell\}_\ell$ and $\mathcal{B}_i = \{B_{i,\ell}\}_\ell$.

At the beginning of communication round $t \in \{1, \dots, T\}$, client i constructs the personalized policy

$$\theta_i^{t-1} = \{W_0, \mathcal{A}^{t-1}, \mathcal{B}_i^{t-1}\}, \quad \pi_i^{t-1} = \pi_{\theta_i^{t-1}}, \quad (3)$$

where $\mathcal{A}^{t-1}$ is broadcast by the server and $\mathcal{B}_i^{t-1}$ is retained locally. The initial LoRA factors induce a near-zero adapter update, so $\pi_{\theta_i^0}$ has the same initial behavior as the frozen backbone. Our objective is to transfer shared preference information through $\mathcal{A}$ while preserving client-specific adaptations in $\mathcal{B}_i$.

3.2 Method Overview

FEDCARE uses the following order in each communication round: candidate scoring, curriculum selection, reference refresh, local DPO, and LoRA-A aggregation. Each client first scores its available preference pairs against the reference retained from the preceding round and constructs a client-adaptive curriculum. Only after selection does DPRE refresh the reference from the current personalized checkpoint. The client then optimizes its local policy against this frozen reference and uploads the updated LoRA-A factor. Round 1 uses the base-policy preference gap for curriculum scoring, as specified in Eq. (5).

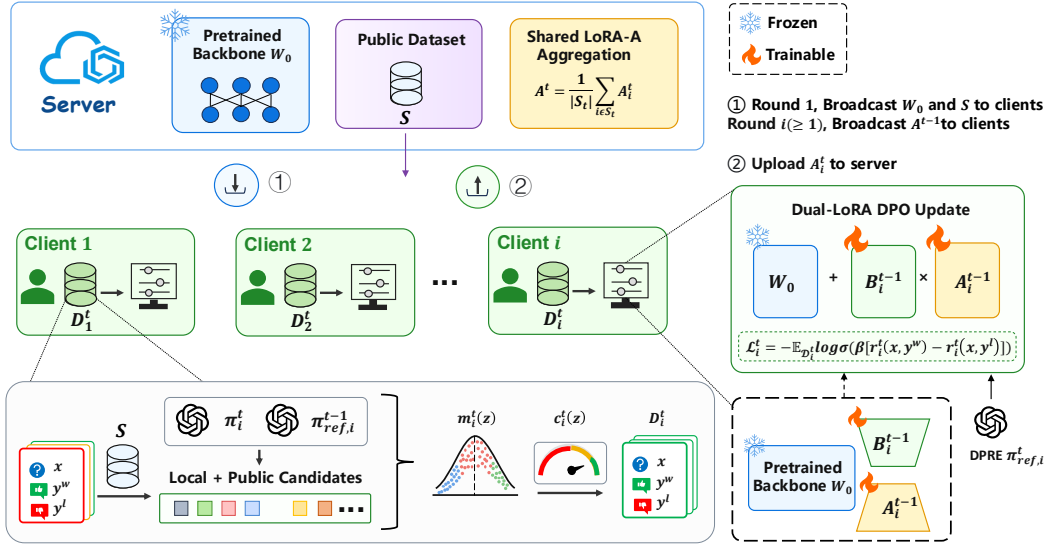


Figure 2: Overview of FEDCARE. In communication round t , the server broadcasts the shared LoRA-A factor A^{t-1} . Each client scores its staged candidate pool using the current personalized policy and the frozen reference retained from the preceding round, then constructs curriculum $\mathcal{D}_i^t$. After selection and before local DPO, DPPE refreshes the reference from the current personalized checkpoint. The reference remains frozen during local optimization. Each client uploads only its updated LoRA-A factor for server aggregation; LoRA-B and local preference data remain client-local.

3.3 Client-Adaptive Curriculum Construction

A fixed curriculum may be unsuitable under client heterogeneity because the same nominal difficulty can produce different learning signals for different client policies. FEDCARE therefore separates candidate availability from candidate ranking. The availability schedule progressively exposes easy, medium, and eligible local hard pairs, whereas ranking is recomputed independently by each client in every round.

For a preference pair $z = (x, y^w, y^l)$, define the sequence-level chosen-rejected log-probability gap under policy π_ϕ as

$$g_\phi(z) = \log \pi_\phi(y^w|x) - \log \pi_\phi(y^l|x). \quad (4)$$

Here, $\pi_\phi(y|x)$ denotes the autoregressive probability of the complete response sequence y conditioned on x .

In Round 1, no preceding personalized checkpoint is available. We therefore use the initial policy gap as a cold-start score. From Round 2 onward, the scoring policy is compared with the checkpoint retained from the preceding round:

$$m_i^t(z) = \begin{cases} \beta g_{\theta_i^0}(z), & t = 1, \\ \beta [g_{\theta_i^{t-1}}(z) - g_{\text{stopgrad}(\theta_i^{t-2})}(z)], & t \geq 2. \end{cases} \quad (5)$$

For $t \geq 2$, Eq. (5) is the difference between the implicit DPO rewards assigned to the preferred and rejected responses. Specifically, defining

$$\widehat{r}_i^t(x, y) = \beta \log \frac{\pi_{\theta_i^{t-1}}(y|x)}{\pi_{\theta_i^{t-2}}(y|x)}, \quad (6)$$

gives

$$m_i^t(z) = \widehat{r}_i^t(x, y^w) - \widehat{r}_i^t(x, y^l). \quad (7)$$

Thus, for $t \geq 2$, $m_i^t(z)$ represents the implicit-reward difference between the chosen and rejected responses and quantifies the temporal change in their sequence-level log-probability gap. This is the same reward-margin form used by BeeS [8], with a client-specific historical checkpoint as reference. A positive margin indicates increasing relative support for the supplied ordering; it does not require $g_{\theta_i^{t-1}}(z) > 0$. For example, a gap changing from -8 to -6 yields a positive margin. The margin therefore measures reference-relative preference improvement, not absolute agreement with the given ordering.

Conditional link to preference validity. Let $q_i(z) = r_i^*(x, y^w) - r_i^*(x, y^l)$ be the clean client-specific reward gap. Under Bradley-Terry preferences, an independent clean comparison agrees with the supplied ordering with probability $p_i(z) = \sigma(q_i(z))$, conditional on the candidate and pre-comparison history. If $m_i^t(z) = a_i^t q_i(z) + e_i^t(z)$, with $a_i^t > 0$ and $|e_i^t(z)| \leq \delta_i^t$ over the candidate pool, then

$$p_i(z) \geq \sigma\left(\frac{m_i^t(z) - \delta_i^t}{a_i^t}\right). \quad (8)$$

Larger margins therefore improve a conditional validity lower bound. This supports margin-based ranking, without establishing calibrated confidence or reasoning correctness. Appendix B provides the derivation, selected-set guarantee, and limitations, including the separate cold-start assumption for Round 1.

We map the margin to a bounded ordinal score:

$$c_i^t(z) = \text{clip}\left(\frac{\text{clip}(m_i^t(z), L, U) - L}{U - L}, \epsilon, 1 - \epsilon\right), \quad (9)$$

where $L < U$ and $0 < \epsilon < 1/2$. The clipping interval $[L, U]$ controls score saturation rather than calibrating preference probabilities. For

the single-margin mapping, changing this interval preserves the ordering of samples that remain in the non-saturated region. A narrow interval may increase tied scores through saturation, whereas a wider interval retains more score distinctions but compresses their numerical range. Its effect on selection therefore depends on saturation and the downstream filtering rule.

Local-priority curriculum allocation. From the stage-available candidates passing the local hard-pair gate, we first select up to N_t highest-scoring local pairs, forming $\mathcal{L}_i^t$. Public pairs passing the q_p -quantile gate of selected local scores form $\mathcal{P}_i^t$; this gate is inactive if $\mathcal{L}_i^t$ is empty. We then set

$$\begin{aligned} \kappa_i^t &= \min\{N_t - |\mathcal{L}_i^t|, \lfloor \eta_t N_t \rfloor\}, \\ \mathcal{D}_i^t &= \text{Dedup}\left(\mathcal{L}_i^t \cup \text{Top}_{\kappa_i^t}(\mathcal{P}_i^t; c_i^t)\right). \end{aligned} \quad (10)$$

Here η_t caps public count relative to the target budget, and Top_k uses seeded tie-breaking. Negative margins remain eligible. Allocation details and score sensitivity are given in Appendix A.

3.4 Dynamic Personalized Reference Evolution

We use *Dynamic Personalized Reference Evolution (DPRE)* to denote an update-after-scoring personalized reference refresh schedule. Its purpose is to separate two temporal roles of the reference: measuring preference changes for curriculum construction and anchoring the subsequent local DPO update. DPRE does not train a separate reference policy.

Let R_i denote the frozen reference checkpoint retained by client i . In Round 1, $R_i = \text{stopgrad}(\theta_i^0)$, and curriculum scoring uses the base-policy preference gap in Eq. (5). At the start of each subsequent round $t \geq 2$, the current personalized checkpoint is $\theta_i^{t-1} = \{W_0, \mathcal{A}^{t-1}, \mathcal{B}_i^{t-1}\}$, whereas the retained reference is

$$R_i = \text{stopgrad}(\theta_i^{t-2}). \quad (11)$$

Candidate scoring therefore compares the current policy with an earlier personalized checkpoint. Refreshing the reference before scoring would make these checkpoints identical and collapse the relative preference margin to zero, removing the temporal comparison used by the selector.

After constructing $\mathcal{D}_i^t$ and before local DPO optimization, client i refreshes the reference:

$$R_i \leftarrow \text{stopgrad}(\theta_i^{t-1}), \quad \pi_{\text{ref},i}^t = \pi_{R_i}. \quad (12)$$

Here, θ_i^{t-1} contains the server-aggregated LoRA-A factor and the client-retained LoRA-B factor. The refreshed reference remains frozen during local DPO, after aggregation, and through candidate scoring in Round $t + 1$; it is refreshed again only after that round's curriculum selection. The scoring reference and the local DPO reference thus correspond to pre-refresh and post-refresh checkpoints, not independently learned policies.

3.5 Dual-LoRA Adaptation and Local DPO

Under non-IID client distributions, aggregating all personalized parameters can overwrite domain-specific adaptations. Following FedSA-LoRA [11], we adopt asymmetric factor sharing: LoRA-A is aggregated across clients, while LoRA-B remains local. We refer to this existing personalization mechanism as Dual-LoRA. Within

Algorithm 1 FEDCARE

Require: Frozen backbone W_0 ; clients $[M]$; number of rounds T ; initial shared LoRA-A factor $\mathcal{A}^0$; local LoRA-B factors $\{\mathcal{B}_i^0\}_{i=1}^M$

- 1: Initialize $\theta_i^0 = \{W_0, \mathcal{A}^0, \mathcal{B}_i^0\}$ and $R_i \leftarrow \text{stopgrad}(\theta_i^0)$ for each client i
- 2: **for** round $t = 1, \dots, T$ **do**
- 3: Server broadcasts $\mathcal{A}^{t-1}$ to clients $\mathcal{S}_t$
- 4: **for** client $i \in \mathcal{S}_t$ in parallel **do**
- 5: Form the current pre-update policy $\theta_i^{t-1} = \{W_0, \mathcal{A}^{t-1}, \mathcal{B}_i^{t-1}\}$
- 6: **if** $t = 1$ **then**
- 7: Score candidates using the initial-policy chosen-rejected log-probability gap
- 8: **else**
- 9: Score candidates using θ_i^{t-1} and the retained reference $R_i = \text{stopgrad}(\theta_i^{t-2})$
- 10: **end if**
- 11: Greedy construct $\mathcal{D}_i^t$ under the selection budget, round-specific score gates, and active public-ratio constraint
- 12: After selection, refresh the reference: $R_i \leftarrow \text{stopgrad}(\theta_i^{t-1})$
- 13: Initialize the trainable local policy: $\vartheta_i^t \leftarrow \theta_i^{t-1}$
- 14: Optimize ϑ_i^t on $\mathcal{D}_i^t$ against the frozen reference π_{R_i} using $\mathcal{L}_i^t$
- 15: Obtain updated factors $(\mathcal{A}_i^t, \mathcal{B}_i^t)$
- 16: Upload $\mathcal{A}_i^t$ and retain $\mathcal{B}_i^t$ locally
- 17: **end for**
- 18: Server aggregates $\mathcal{A}^t = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \mathcal{A}_i^t$
- 19: **for** client $i \in \mathcal{S}_t$ **do**
- 20: Form the next-round personalized checkpoint $\theta_i^t = \{W_0, \mathcal{A}^t, \mathcal{B}_i^t\}$
- 21: Keep $R_i = \text{stopgrad}(\theta_i^{t-1})$ frozen until the next-round curriculum has been constructed
- 22: **end for**
- 23: **end for**
- 24: **return** $\{\pi_{\vartheta_i^T}\}_{i=1}^M$

FEDCARE, it provides the client-specific policy checkpoints used for curriculum scoring and personalized reference evolution.

After curriculum selection and reference refresh, client i initializes the trainable policy as $\vartheta_i^t \leftarrow \theta_i^{t-1}$. For a preference pair $z = (x, y^w, y^l)$, define

$$\Delta_i^t(z) = \log \frac{\pi_{\vartheta_i^t}(y^w|x)}{\pi_{R_i}(y^w|x)} - \log \frac{\pi_{\vartheta_i^t}(y^l|x)}{\pi_{R_i}(y^l|x)}. \quad (13)$$

The local DPO objective is

$$\mathcal{L}_i^t = -\mathbb{E}_{z \sim \mathcal{D}_i^t} [\log \sigma(\beta \Delta_i^t(z))], \quad (14)$$

where $\beta > 0$ scales the reference-relative preference gap and the expectation averages uniformly over the selected pairs. Local optimization increases the chosen-response advantage over the rejected response relative to the frozen reference. The backbone W_0 remains fixed, and only the trainable policy's LoRA factors are updated, yielding $(\mathcal{A}_i^t, \mathcal{B}_i^t)$.

Local optimization interpretation. Both the selected curriculum $\mathcal{D}_i^t$ and the refreshed reference R_i remain fixed during local optimization, providing a fixed within-round training objective. At initialization, $\theta_i^t = \theta_i^{t-1}$ and $R_i = \text{stopgrad}(\theta_i^{t-1})$. The policy and reference therefore represent the same distribution, giving $\Delta_i^t(z) = 0$. For the per-pair loss $\ell_i^t(z) = -\log \sigma(\beta \Delta_i^t(z))$,

$$\begin{aligned} \frac{\partial \ell_i^t(z)}{\partial \Delta_i^t(z)} &= -\beta \sigma(-\beta \Delta_i^t(z)), \\ \left. \frac{\partial \ell_i^t(z)}{\partial \Delta_i^t(z)} \right|_{\Delta_i^t(z)=0} &= -\frac{\beta}{2}. \end{aligned} \quad (15)$$

Reference refresh thus centers the initial DPO logit without eliminating its scalar learning signal. The reference receives no gradients, while policy parameter gradients additionally depend on the derivative of the chosen-rejected log-probability gap.

More generally, the loss satisfies

$$\left| \frac{\partial \ell_i^t(z)}{\partial \Delta_i^t(z)} \right| \leq \beta, \quad 0 \leq \frac{\partial^2 \ell_i^t(z)}{\partial (\Delta_i^t(z))^2} \leq \frac{\beta^2}{4}. \quad (16)$$

These bounds characterize the loss response to the reference-relative gap. Together with the frozen reference, they clarify the within-round optimization behavior, without implying parameter-space convexity or a hard bound on policy drift.

Effect of selection on local gradients. The conditional validity analysis connects curriculum selection to the local DPO update. For a nonempty selected curriculum, let $\gamma_i^t = \min_{z \in \mathcal{D}_i^t} m_i^t(z)$ be its minimum raw scoring margin. Let $C_z = 1$ indicate that an independent clean comparison agrees with the supplied ordering, and let $\mathcal{H}_i^t$ denote the pre-comparison history fixing the selected set. Under the assumptions of Eq. (8), the expected disagreement fraction satisfies

$$\mathbb{E} \left[\frac{\sum_{z \in \mathcal{D}_i^t} (1 - C_z)}{|\mathcal{D}_i^t|} \middle| \mathcal{H}_i^t \right] \leq \sigma \left(\frac{\delta_i^t - \gamma_i^t}{a_i^t} \right). \quad (17)$$

For the local training logit $h_\vartheta(z) = \beta \Delta_i^t(z)$, define the clean-comparison loss as $\ell_c = -C \log \sigma(h_\vartheta) - (1 - C) \log \sigma(-h_\vartheta)$. Let $\mathcal{L}_{\text{clean},i}^t$ average this loss over the same selected pairs and independent clean comparison outcomes, using the same frozen reference. At a fixed local parameter value ϑ , suppose $\|\nabla_\vartheta \Delta_i^t(z)\| \leq G_i^t$ for every selected pair. The discrepancy between the supplied-label and expected clean-label gradients then satisfies

$$\begin{aligned} \left\| \nabla_\vartheta \mathcal{L}_i^t - \nabla_\vartheta \mathcal{L}_{\text{clean},i}^t \right\| \\ \leq \beta G_i^t \sigma \left(\frac{\delta_i^t - \gamma_i^t}{a_i^t} \right). \end{aligned} \quad (18)$$

A larger minimum selected margin therefore tightens this gradient-discrepancy bound under the stated assumptions. This provides an optimization-level rationale for margin-based selection, complementing DPPE's fixed within-round reference. The result concerns the realized curriculum and does not establish global federated convergence. The derivation is provided in Appendix B.

3.6 LoRA-A Aggregation

After local optimization, client i uploads only $\mathcal{A}_i^t$ and retains $\mathcal{B}_i^t$ locally. Given the participating clients $\mathcal{S}_t$, the server computes

$$\mathcal{A}^t = \frac{1}{|\mathcal{S}_t|} \sum_{i \in \mathcal{S}_t} \mathcal{A}_i^t. \quad (19)$$

We use uniform client averaging, assigning equal weight to each participating client irrespective of its realized curriculum size. The personalized checkpoint used by client i in the next communication round is

$$\theta_i^t = \{W_0, \mathcal{A}^t, \mathcal{B}_i^t\}. \quad (20)$$

The shared LoRA-A factor transfers information across clients, whereas the local LoRA-B factor preserves client-specific adaptation. Meanwhile, the reference remains frozen at $\text{stopgrad}(\theta_i^{t-1})$. Round $t+1$ therefore scores candidates using θ_i^t against θ_i^{t-1} before refreshing the reference for the next local DPO update.

4 Experiment

4.1 Experimental Setup

Datasets and Federated Partitions. We evaluate FEDCARE on heterogeneous mathematical reasoning and code-to-text generation benchmarks. The main mathematical reasoning experiments use Hendrycks MATH [13], where different mathematical subjects are treated as different clients. The code-to-text experiments use CodeXGLUE code-to-text [16], where different programming languages correspond to different clients. We additionally use MetaMathQA [41] as the public mathematical preference pool for curriculum construction. To further examine robustness, we consider additional evaluation settings with GSM8K [6] and programming challenge generation on APPS [12]. For client-number scalability, we construct non-IID GSM8K partitions with different numbers of clients and Dirichlet concentration parameters.

Baselines and Evaluation Metrics. We compare FEDCARE with representative centralized, local, and federated preference-optimization baselines, including the base model, Local DPO, FedIT[39], FedVA[39], and FedSA-LoRA[11]. These baselines cover non-federated local adaptation, standard federated instruction tuning, federated value alignment, and selective LoRA aggregation. We use task-native evaluation metrics: Pass@1 for MATH, GSM8K and APPS, and BLEU for CodeXGLUE code-to-text generation. Unless otherwise specified, we report the macro average over clients or tasks.

Training Configuration. All experiments are conducted using four NVIDIA RTX 6000 GPUs with DeepSpeed. Our training pipeline is implemented based on OpenRLHF [15]. Qwen2.5-7B [34, 37] is used as the default backbone, while Qwen2.5-Math-7B [38], Llama-3.2-3B [10], and Mistral-7B [17] are used to evaluate backbone generality. Qwen2.5-7B-Instruct is used during initial preference-data construction, and mathematical reasoning is evaluated with the official Qwen2.5-Math evaluation framework. All LoRA-based methods use rank $r = 64$, scaling factor $\alpha = 128$, and dropout 0.1. Complete training and selection hyperparameters are provided in the appendix.

Table 1: Performance comparison on heterogeneous federated mathematical reasoning tasks. All results are evaluated with pass@1 using one greedily decoded response per problem with temperature 0. Centralized denotes a pooled-data DPO reference run, rather than an upper bound, because centralized and federated optimization follow different training topologies. Bold indicates the best result, and underline indicates the second-best result; the same notation is used in the following tables.

Task	Centralized	Base	Local DPO	FedIT	FedVA	FedSA-LoRA	Ours
Algebra	87.90	80.50	<u>89.30</u>	86.40	86.00	<u>89.30</u>	91.20
Counting	67.10	50.80	63.70	60.50	57.40	<u>66.70</u>	69.60
Geometry	54.50	49.70	<u>58.50</u>	56.20	52.60	56.40	60.80
Inter. Alg.	50.70	47.20	54.40	50.70	51.60	51.50	<u>53.20</u>
Num. Theory	70.00	62.40	<u>73.90</u>	66.30	69.60	71.50	74.40
Prealgebra	84.00	72.00	<u>81.70</u>	79.70	80.10	80.10	86.20
Precalc.	57.00	52.60	57.00	54.20	55.90	55.30	<u>56.40</u>
Avg.	67.31	59.31	<u>68.36</u>	64.86	64.74	67.26	70.26

Selection and Reporting Protocol. FEDCARE selects curriculum data by greedily ranking eligible preference pairs under the round-specific budget and public-data constraints. To ensure a controlled comparison, we fix the seeds used for tie-breaking and data ordering across methods; Random Subset is the only variant in which candidate membership is sampled randomly. During evaluation, all models use greedy decoding with temperature 0, making the reported scores deterministic for a fixed trained checkpoint. Unless otherwise specified, we report checkpoint-level results under this fixed protocol, and all comparisons are interpreted under the same training budget, data split, and evaluation setting.

4.2 Overall Performance

Performance Analysis. Tables 1 and 2 show that FEDCARE improves federated preference optimization across both mathematical reasoning and code-to-text generation. On MATH, our method achieves the best average accuracy of 70.26%, exceeding the strongest baseline by 1.90 accuracy points. On CodeXGLUE, our method also obtains the highest average BLEU score of 24.24, exceeding the strongest baseline by 3.47 BLEU. These results indicate that client-adaptive curriculum construction, Dual-LoRA personalization, and DPRE are effective across different heterogeneous tasks.

Analysis of Curriculum Evolution. Figure 3 illustrates the selected curriculum across communication rounds on MATH. Candidate availability follows a staged expansion from easy to medium and eligible local hard pairs. Within each stage, however, clients greedily reselect samples according to their own margin distributions and reliability gates. The adaptive component therefore lies in client-specific within-stage selection, rather than in learning the difficulty-availability schedule itself.

Analysis of Communication-Round Dynamics. Figure 4 shows the performance gain over the initial model under the same communication budget. After three rounds, FEDCARE achieves gains of 10.95 accuracy points on MATH and 12.96 BLEU on CodeXGLUE, exceeding the gains of the compared baselines. These results demonstrate the effectiveness of FEDCARE in improving mathematical reasoning and code-to-text generation over the evaluated communication rounds. The favorable gains under a fixed communication

Table 2: Performance comparison on heterogeneous federated code-to-text tasks. BLEU-4 on the CodeXGLUE code-to-text test split. Each client is evaluated on its own programming language.

Lang.	Base	Local	FedIT	FedVA	FedSA-LoRA	Ours
Go	6.02	<u>9.12</u>	5.48	5.52	8.58	12.06
Java	7.57	<u>10.49</u>	9.62	9.62	9.89	11.06
JS	5.60	<u>7.94</u>	6.23	6.26	7.37	8.53
PHP	10.27	16.12	13.74	13.76	<u>15.42</u>	14.68
Py.	32.95	74.61	48.35	47.92	<u>73.37</u>	90.81
Ruby	5.28	6.33	6.18	6.44	5.97	8.27
Avg.	11.28	<u>20.77</u>	14.93	14.92	20.10	24.24

Table 3: Cross-domain generalization under heterogeneous federated clients. All tasks are evaluated using Pass@1, and all results are reported as percentages.

Task	Base	Local	FedIT	FedVA	FedSA	Ours
GSM8K	58.78	<u>85.90</u>	81.30	79.10	83.50	86.20
MATH	62.10	72.60	64.60	67.40	71.50	<u>72.40</u>
APPS	3.94	7.38	4.80	8.78	7.80	<u>8.10</u>
Avg.	41.60	<u>55.29</u>	50.23	51.76	54.27	55.57

budget support the overall framework, while the component ablations further examine the contributions of curriculum selection, personalization, and reference evolution.

Separate Cross-Domain Protocol. Table 3 is obtained from a separate cross-domain federation, in which GSM8K, MATH, and APPS constitute different task-domain clients. It therefore uses a different client partition and training pool from the subject-partitioned MATH setting in Table 1; the two MATH scores should not be interpreted as repeated measurements under an identical protocol.

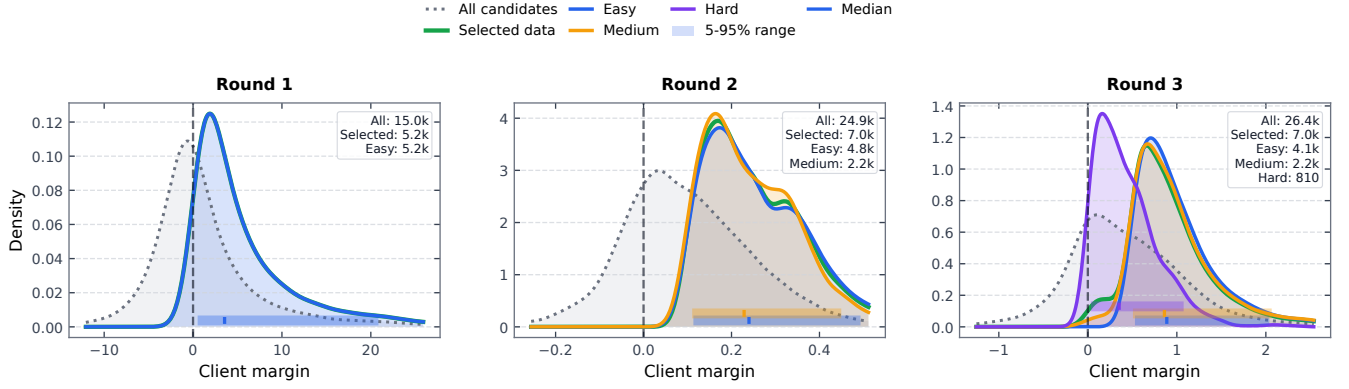


Figure 3: Model-aware curriculum selection on MATH across three communication rounds. Each panel shows the client-margin distribution of all candidates and the selected samples, grouped by difficulty.

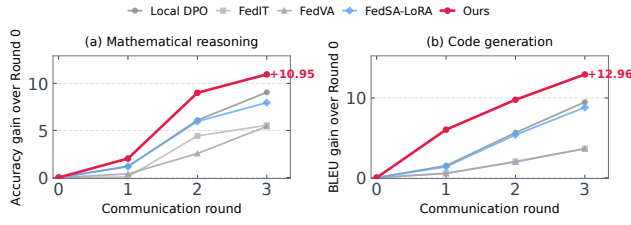


Figure 4: Performance gains under a fixed communication budget. Round 0 denotes the initial model before federated preference optimization.

Table 4: Performance with different backbone models. Each backbone is trained in a separate federated run.

Backbone	Size	Base	FedIT	FedVA	FedSA	Ours
Qwen2.5	7B	59.31	64.86	64.74	67.26	70.26
Qwen-2.5-Math	7B	51.53	56.18	54.56	55.07	56.87
Llama-3.2	3B	2.41	4.17	4.71	4.86	5.46
Mistral	7B	0.46	5.10	3.57	4.47	8.03

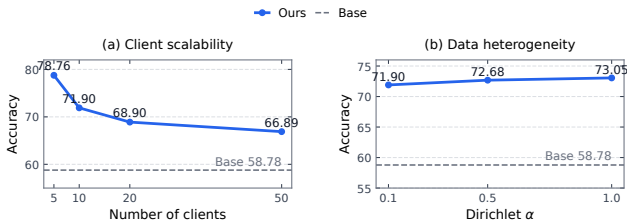


Figure 5: Scalability of FEDCARE on GSM8K. Smaller α indicates stronger client heterogeneity.

4.3 Generalization and Robustness Analysis

Analysis of Cross-Domain Generalization. Table 3 reports Pass@1 performance under cross-domain federated heterogeneity. FEDCARE obtains the highest macro average in this protocol. However, its margin over Local DPO is small, and it does not achieve the best score on every individual task. We therefore interpret this result as evidence of a favorable aggregate trade-off under this federation, rather than uniform superiority across domains.

Analysis of Backbone Robustness. Table 4 shows that FEDCARE consistently improves performance across different backbone models. The gains appear on Qwen2.5, Qwen2.5-Math, Llama-3.2, and Mistral, indicating that the proposed method is not tied to a specific base model family. Although the absolute performance varies with the backbone capability, the consistent improvement suggests that client-adaptive curriculum selection remains beneficial under different initialization strengths.

Analysis of Client Scalability. Figure 5 evaluates FEDCARE with increasing client counts and different Dirichlet heterogeneity levels. As the federation grows, accuracy decreases from 78.76% to 66.89% in Figure 5(a), highlighting the challenges of learning across a larger federation under the evaluated partitioning protocol. Nevertheless, FEDCARE remains applicable across the tested client scales, using client-specific curriculum selection and selective LoRA sharing without pooling client preference data. These results support its applicability to larger federations while identifying performance degradation with increasing scale as an area for further improvement.

4.4 Ablation Study

Analysis of Aggregation Rules. Table 5 compares single-run point estimates obtained with different server-side aggregation rules. Their macro averages are close: FEDCARE with FedAvg obtains 70.26%, compared with 70.16% for FedProx, 69.34% for FedAdam, and 69.00% for FedYogi. The 0.10-point difference between FedAvg and FedProx is too small to support a claim that one aggregation rule is statistically superior. Instead, the result shows that the proposed framework does not require a complex server optimizer to obtain

Table 5: Comparison of federated optimization variants on MATH. FedProx adds a local proximal penalty on the shared LoRA-A factors, whereas FedAdam and FedYogi modify server-side updates. Ours uses uniform averaging of LoRA-A. The remaining FedCARE configuration is held fixed.

Task	FedProx	FedAdam	FedYogi	Ours
Algebra	<u>90.70</u>	90.10	90.10	91.20
Counting	<u>66.90</u>	66.90	66.20	69.60
Geometry	<u>60.10</u>	57.80	58.50	60.80
Inter. Alg.	<u>54.70</u>	54.80	54.00	53.20
Num. Theory	75.40	73.50	73.70	<u>74.40</u>
Prealgebra	86.70	84.80	83.50	<u>86.20</u>
Precalc.	56.60	57.50	<u>57.00</u>	56.40
Avg.	<u>70.16</u>	69.34	69.00	70.26

Table 6: Component ablations on MATH and CodeXGLUE. Δ denotes the change relative to FedCARE; blue values indicate decreases.

Variant	MATH		CodeXGLUE	
	Avg. Acc.	Δ	Avg. BLEU	Δ
<i>w/o Data Selection</i>				
Full Data	67.91	-2.35	23.67	-0.57
<i>w/o Client-Adaptive Curriculum Selection</i>				
Random Subset	66.77	-3.49	21.10	-3.14
Fixed Curriculum	66.86	-3.40	21.24	-3.00
w/o Dual-LoRA	67.23	-3.03	15.70	-8.54
w/o DPPE	69.16	-1.10	20.13	-4.11
Ours	70.26	–	24.24	–

Method	Shared update	Local state	Upload / round	Time / round
Full Data	Full LoRA	–	161.48M	3h14m
FedIT	Full LoRA	–	161.48M	1h49m
FedVA	Full LoRA	–	161.48M	2h02m
FedSA	LoRA-A	LoRA-B	72.48M	1h58m
Ours	LoRA-A	LoRA-B + ref.	72.48M	2h14m

Table 7: Efficiency and communication comparison. Full Data uses the complete preference pool. FedSA and Ours reduce communication by uploading only LoRA-A, while Ours adds curriculum selection and reference evolution.

its reported performance under this protocol. We therefore use FedAvg as the simplest aggregation rule for the main experiments.

Analysis of Key Components. Table 6 examines the contributions of client-adaptive curriculum selection, Dual-LoRA personalization, and DPPE. The complete framework achieves the highest average performance on both benchmarks, and removing any component

reduces its performance. Random Subset and Fixed Curriculum underperform FedCARE, supporting the benefit of adapting sample selection to each client’s evolving policy state. Removing Dual-LoRA reduces MATH accuracy by 3.03 points and CodeXGLUE BLEU by 8.54, highlighting the importance of personalized adaptation. Removing DPPE yields corresponding decreases of 1.10 accuracy points and 4.11 BLEU, supporting the utility of the personalized reference-refresh schedule. Together, these results support integrating adaptive selection, personalization, and reference evolution within federated preference optimization.

Analysis of Efficiency and Communication. Table 7 compares the computation and communication costs of different training strategies. Full Data trains on the complete available preference pool without curriculum filtering, which leads to the longest per-round runtime and requires uploading the full LoRA adapter. In contrast, FedCARE uses model-aware curriculum selection to train on a smaller but more reliable subset, reducing the per-round training time from 3h14m to 2h14m. Meanwhile, it uploads only LoRA-A and keeps LoRA-B client-local, resulting in the same communication cost as FedSA and much lower upload size than full-LoRA aggregation. This shows that the proposed curriculum improves data efficiency while preserving communication efficiency.

4.5 Privacy Considerations

FedCARE keeps preference data, personalized LoRA-B factors, and reference checkpoints local, communicating only LoRA-A factors for aggregation. This reduces direct exposure of client-specific information, although shared updates may still reveal information about local data and do not provide a formal privacy guarantee. All experiments use public datasets with simulated clients. Additional safeguards, such as differential privacy or secure aggregation, could be incorporated for privacy-sensitive deployments.

5 Conclusion

This paper introduces FedCARE, a federated DPO framework for LLMs under heterogeneous client distributions. FedCARE builds client-adaptive curriculum by selecting preference pairs according to client’s evolving policy state, shares transferable LoRA-A factors while keeping client-specific LoRA-B factors local, and refreshes the personalized reference policy through DPPE. Experiments on mathematical reasoning and code-to-text generation benchmarks show that FedCARE achieves the best average performance among the compared methods, remains effective under cross-domain heterogeneity, and works well with simple FedAvg aggregation. These results suggest that effective federated preference optimization requires not only communication-efficient parameter sharing, but also adaptive control over the preference signals introduced to each client.

References

- [1] Manoj Ghuhan Arivazhagan, Vinay Aggarwal, Aaditya Kumar Singh, and Sunav Choudhary. 2019. Federated Learning with Personalization Layers. *arXiv:1912.00818 [cs.LG]* <https://arxiv.org/abs/1912.00818>
- [2] Mohammad Gheshlaghi Azar, Zhaohan Daniel Guo, Bilal Piot, Remi Munos, Mark Rowland, Michal Valko, and Daniele Calandriello. 2024. A general theoretical paradigm to understand learning from human preferences. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 4447–4455.
- [3] Jiamu Bai, Daoyuan Chen, Bingchen Qian, Liuyi Yao, and Yaliang Li. 2024. Federated fine-tuning of large language models under heterogeneous tasks and client resources. *Advances in Neural Information Processing Systems* 37 (2024), 14457–14483.
- [4] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. Curriculum learning. In *Proceedings of the 26th annual international conference on machine learning*. 41–48.
- [5] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. *Advances in neural information processing systems* 30 (2017).
- [6] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. *arXiv:2110.14168 [cs.LG]* <https://arxiv.org/abs/2110.14168>
- [7] Liam Collins, Hamed Hassani, Aryan Mokhtari, and Sanjay Shakkottai. 2021. Exploiting shared representations for personalized federated learning. In *International conference on machine learning*. PMLR, 2089–2099.
- [8] Xun Deng, Han Zhong, Rui Ai, Fuli Feng, Zheng Wang, and Xiangnan He. 2025. Less is more: Improving llm alignment via preference data selection. *Advances in Neural Information Processing Systems* 38 (2025), 161259–161285.
- [9] Kavin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. 2024. KTO: Model Alignment as Prospect Theoretic Optimization. *arXiv:2402.01306 [cs.LG]* <https://arxiv.org/abs/2402.01306>
- [10] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The Llama 3 Herd of Models. *arXiv:2407.21783 [cs.AI]* <https://arxiv.org/abs/2407.21783>
- [11] Pengxin Guo, Shuang Zeng, Yanran Wang, Huijie Fan, Feifei Wang, and Liangqiong Qu. 2025. Selective aggregation for low-rank adaptation in federated learning. In *International Conference on Learning Representations*, Vol. 2025. 99003–99027.
- [12] Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Song, and Jacob Steinhardt. 2021. Measuring Coding Challenge Competence With APPS. *NeurIPS* (2021).
- [13] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring Mathematical Problem Solving With the MATH Dataset. *NeurIPS* (2021).
- [14] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. 2022. Lora: Low-rank adaptation of large language models. *Iclr* 1, 2 (2022), 3.
- [15] Jian Hu, Xibin Wu, Wei Shen, Jason Klein Liu, Zilin Zhu, Weixun Wang, Songlin Jiang, Haoran Wang, Hao Chen, Bin Chen, Weikai Fang, Xianyu, Yu Cao, Haotian Xu, and Yiming Liu. 2025. OpenRLHF: An Easy-to-use, Scalable and High-performance RLHF Framework. *arXiv:2405.11143 [cs.AI]* <https://arxiv.org/abs/2405.11143>
- [16] Hamel Husain, Ho-Hsiang Wu, Tiferet Gazit, Miltiadis Allamanis, and Marc Brockschmidt. 2020. CodeSearchNet Challenge: Evaluating the State of Semantic Code Search. *arXiv:1909.09436 [cs.LG]* <https://arxiv.org/abs/1909.09436>
- [17] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Léo Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7B. *arXiv:2310.06825 [cs.CL]* <https://arxiv.org/abs/2310.06825>
- [18] Peter Kairouz and H Brendan McMahan. 2021. Advances and open problems in federated learning. *Foundations and trends in machine learning* 14, 1-2 (2021), 1–210.
- [19] M Kumar, Benjamin Packer, and Daphne Koller. 2010. Self-paced learning for latent variable models. *Advances in neural information processing systems* 23 (2010).
- [20] Tian Li, Anit Kumar Sahu, Ameet Talwalkar, and Virginia Smith. 2020. Federated learning: Challenges, methods, and future directions. *IEEE signal processing magazine* 37, 3 (2020), 50–60.
- [21] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020. Federated optimization in heterogeneous networks. *Proceedings of Machine learning and systems* 2 (2020), 429–450.
- [22] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let’s verify step by step. In *International Conference on Learning Representations*, Vol. 2024. 39578–39601.
- [23] Ji Liu, Jiaxiang Ren, Ruoming Jin, Zijie Zhang, Yang Zhou, Patrick Valduriez, and Dejing Dou. 2024. Fisher information-based efficient curriculum federated learning with large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 10497–10523.
- [24] Shunyu Liu, Wenkai Fang, Zetian Hu, Junjie Zhang, Yang Zhou, Kongcheng Zhang, Rongcheng Tu, Ting-En Lin, Fei Huang, Mingli Song, Yongbin Li, and Dacheng Tao. 2025. A Survey of Direct Preference Optimization. *arXiv:2503.11701 [cs.LG]* <https://arxiv.org/abs/2503.11701>
- [25] Yao Lu, Jiaxing Qi, Zhongzhi Luan, Shaohan Huang, Carol Fung, Hailong Yang, and Depei Qian. 2026. FDLora: Personalized Federated Learning of Large Language Model via Dual LoRA Tuning. *arXiv:2406.07925 [cs.DC]* <https://arxiv.org/abs/2406.07925>
- [26] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. Pmlr, 1273–1282.
- [27] Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. Simpo: Simple preference optimization with a reference-free reward. *Advances in Neural Information Processing Systems* 37 (2024), 124198–124235.
- [28] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in neural information processing systems* 35 (2022), 27730–27744.
- [29] Pulkit Pattnaik, Rishabh Maheshwary, Kelechi Ogueji, Vikas Yadav, and Sathwik Tejaswi Madhusudan. 2024. Enhancing alignment using curriculum learning & ranked preferences. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 12891–12907.
- [30] Emmanouil Antonios Platanios, Otilia Stretcu, Graham Neubig, Barnabas Poczos, and Tom Mitchell. 2019. Competence-based curriculum learning for neural machine translation. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 1162–1172.
- [31] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems* 36 (2023), 53728–53741.
- [32] Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F Christiano. 2020. Learning to summarize with human feedback. *Advances in neural information processing systems* 33 (2020), 3008–3021.
- [33] Alysa Ziyang Tan, Han Yu, Lizhen Cui, and Qiang Yang. 2023. Towards personalized federated learning. *IEEE transactions on neural networks and learning systems* 34, 12 (2023), 9587–9603.
- [34] Qwen Team. 2024. Qwen2.5: A Party of Foundation Models. <https://qwenlm.github.io/blog/qwen2.5/>
- [35] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. *arXiv:2203.11171 [cs.CL]* <https://arxiv.org/abs/2203.11171>
- [36] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.
- [37] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang, et al. 2024. Qwen2 Technical Report. *arXiv:2407.10671 [cs.CL]* <https://arxiv.org/abs/2407.10671>
- [38] An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. 2024. Qwen2.5-Math Technical Report: Toward Mathematical Expert Model via Self-Improvement. *arXiv:2409.12122 [cs.CL]* <https://arxiv.org/abs/2409.12122>
- [39] Rui Ye, Wenhao Wang, Jingyi Chai, Dihan Li, Zexi Li, Yinda Xu, Yaxin Du, Yanfeng Wang, and Siheng Chen. 2024. Openfedllm: Training large language models on decentralized private data via federated learning. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining*. 6137–6147.
- [40] Liping Yi, Han Yu, Gang Wang, Xiaoguang Liu, and Xiaoxiao Li. 2024. pFedLoRA: Model-Heterogeneous Personalized Federated Learning with LoRA Tuning. *arXiv:2310.13283 [cs.LG]* <https://arxiv.org/abs/2310.13283>
- [41] Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T. Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. 2024. MetaMath: Bootstrap Your Own Mathematical Questions for Large Language Models. *arXiv:2309.12284 [cs.CL]* <https://arxiv.org/abs/2309.12284>

A Implementation and Selection Details

Candidate admission. The three-round availability schedule is specified in the main paper. In Round 3, let $\mathcal{G}_i^t$ contain the scores of valid local easy and medium candidates. A local hard pair is admitted when

$$c_i^t(z) \geq Q_{0.5}(\mathcal{G}_i^t). \quad (21)$$

The hard gate is inactive if $\mathcal{G}_i^t$ is empty. After local ranking, public candidates must reach the 25th percentile of selected local scores; this gate is inactive if no local pair is selected. Quantiles use the sorted value at zero-based index $\text{round}((n-1)q)$. Ties use seeded random ordering. Negative margins are not separately excluded. Deduplication identifies pairs by their prompt, chosen response, and rejected response; it occurs after allocation and can reduce the final curriculum size.

Public budget and parameter rationale. Equation (10) prioritizes local pairs and fills only the residual budget with eligible public pairs. The public-count cap is $\lfloor \eta_t N_t \rfloor$, with $\eta_1 = 0.5$ and $\eta_2 = \eta_3 = 1$; it is not a bound on the public fraction of the realized set. The Round 1 cap limits supplementation during cold start. Later rounds retain local-first allocation and the relative admission gate without an additional public-count cap. The lower-quartile gate provides a permissive client-relative screen rather than a fixed absolute threshold. The symmetric interval $[-0.5, 0.5]$ maps zero margin to 0.5, and $\epsilon = 10^{-4}$ avoids exact endpoint scores. These are design choices, not empirically optimized thresholds.

Single-field score sensitivity. For the mapping f in Eq. (9), nonexpansiveness of clipping gives

$$|f(m+v) - f(m)| \leq \frac{|v|}{U-L}. \quad (22)$$

If each margin changes by at most v , each score changes by at most $s = v/(U-L)$. On a fixed candidate list, a top- k boundary gap greater than $2s$ preserves membership. A quantile changes by at most s , so a gate decision is preserved when its original threshold distance exceeds $2s$, provided the reference list is fixed. Changing $[L, U]$ preserves rankings and quantile decisions when all compared scores remain in the strictly affine region under both intervals. Saturation and changing candidate lists can invalidate this conclusion; these bounds do not establish empirical insensitivity.

Multi-field implementation. The released selector also supports pooling available margin fields. For field j , set $p_j = f(m_j)$ and compute

$$c_{\text{pool}} = \frac{\prod_j p_j}{\prod_j p_j + \prod_j (1-p_j)}. \quad (23)$$

The default field list is margin, client_margin, reward_gap, implicit_reward_gap, teacher_margin and dpo_margin; only finite available values are used. The scorer writes the same value to client_margin and dpo_margin. If these are the only fields, pooling gives $p^2/[p^2 + (1-p)^2]$, a strictly increasing function of the single-field score p . Different additional fields can change rankings. Setting BEES_MARGIN_KEYS=client_margin recovers the single-field rule. The preceding sensitivity bound applies to f ; repeated fields are not independent evidence of preference validity.

B Derivations for Margin-Based Selection

Sufficient condition for margin alignment. Use the main-paper definitions of $q_i(z)$, $p_i(z)$, and $\mathcal{H}_i^t$. For a fixed reference ρ , the ideal KL-regularized reward objective has optimizer $\pi^*(y|x) = \rho(y|x) \exp(ar_i^*(x, y)/\beta)/Z(x)$, assuming a finite normalizer. This follows by writing the objective as $\beta \log Z(x) - \beta \text{KL}(\pi||\pi^*)$. For $t \geq 2$, suppose the current scoring policy π_i^t and retained reference ρ_i^t satisfy, on candidate responses,

$$\log \frac{\pi_i^t(y|x)}{\rho_i^t(y|x)} = \frac{a_i^t r_i^*(x, y)}{\beta} - \log Z_i^t(x) + \xi_i^t(x, y), \quad (24)$$

where $a_i^t > 0$ and $|\xi_i^t(x, y)| \leq b_i^t$. Subtracting chosen and rejected terms cancels the normalizer, yielding

$$\begin{aligned} m_i^t(z) &= a_i^t q_i(z) + e_i^t(z), \\ e_i^t(z) &= \beta[\xi_i^t(x, y^w) - \xi_i^t(x, y^l)], \end{aligned} \quad (25)$$

with $|e_i^t(z)| \leq 2\beta b_i^t$. This is a sufficient condition, not a guarantee of finite local training or aggregation. Round 1 requires a separate alignment assumption because it uses the initial-policy gap.

Validity and selected-set bounds. Under $|e_i^t(z)| \leq \delta_i^t$, $q_i(z) \geq (m_i^t(z) - \delta_i^t)/a_i^t$. Sigmoid monotonicity proves Eq. (8). For two candidates,

$$q_i(z_1) - q_i(z_2) \geq \frac{m_i^t(z_1) - m_i^t(z_2) - 2\delta_i^t}{a_i^t}, \quad (26)$$

so margin separation exceeding $2\delta_i^t$ guarantees their clean-reward ordering. For the realized nonempty curriculum $\mathcal{S} = \mathcal{D}_i^t$ with minimum raw margin γ_i^t , linearity of conditional expectation gives

$$\begin{aligned} \mathbb{E} \left[\frac{\sum_{z \in \mathcal{S}} (1 - C_z)}{|\mathcal{S}|} \middle| \mathcal{H}_i^t \right] &= \frac{\sum_{z \in \mathcal{S}} (1 - p_i(z))}{|\mathcal{S}|} \\ &\leq \sigma \left(\frac{\delta_i^t - \gamma_i^t}{a_i^t} \right). \end{aligned} \quad (27)$$

Mutual independence among comparisons is unnecessary. The budget does not ensure $\gamma_i^t > \delta_i^t$.

Gradient bound. For $h_\theta(z) = \beta \Delta_i^t(z)$, the binary comparison loss has gradient $(\sigma(h_\theta(z)) - C) \nabla_\theta h_\theta(z)$. Consequently,

$$\begin{aligned} \nabla_\theta \ell_1 - \mathbb{E}[\nabla_\theta \ell_{C_z} | z, \mathcal{H}_i^t] \\ = -(1 - p_i(z)) \nabla_\theta h_\theta(z). \end{aligned} \quad (28)$$

Averaging over $\mathcal{S}$, using $\|\nabla_\theta \Delta_i^t(z)\| \leq G_i^t$ and Eq. (27) proves the main-paper gradient-discrepancy bound. Both objectives use the same selected pairs and frozen reference. These conclusions depend on alignment and do not establish calibration or reasoning correctness; sequence margins can also depend on response length and formatting.

Conditional local descent. View Eq. (14) as a function of the trainable LoRA factors ϕ . If its gradient is L_F -Lipschitz along a full-gradient step $\phi^+ = \phi - \lambda \nabla \mathcal{L}_i^t(\phi)$, the descent lemma gives

$$\begin{aligned} \mathcal{L}_i^t(\phi^+) &\leq \mathcal{L}_i^t(\phi) \\ &\quad - \lambda \left(1 - \frac{L_F \lambda}{2} \right) \|\nabla \mathcal{L}_i^t(\phi)\|^2, \end{aligned} \quad (29)$$

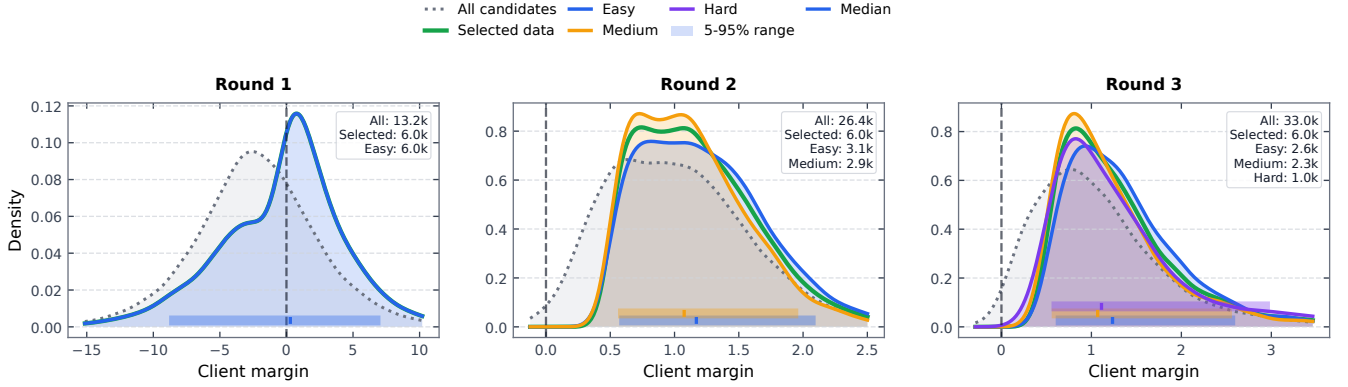


Figure 6: Candidate and selected margin distributions on CodeXGLUE, grouped by client, difficulty, and communication round.

Table 8: Training and selection settings. N_t is a target budget; the realized size may be smaller.

Item	Setting
Default backbone	Qwen2.5-7B
Hardware	4 NVIDIA RTX 6000 GPUs
Clients	7 MATH; 6 CodeXGLUE
Communication rounds	3
DPO scale β	0.1
Score implementation	Available-field pooling
Clipping $[L, U]$; floor ϵ	$[-0.5, 0.5]$; 10^{-4}
Budget $N_1 = N_2 = N_3$	1000 pairs per client
Public cap $\eta_1; \eta_{2,3}$	0.5; 1 (no additional cap)
Public / hard quantile	0.25 / 0.50 (Round 3)
Negative-margin exclusion	None
Tie-breaking / ordering	Seeded
Maximum sequence length	1024
Learning rate / warm-up	2×10^{-6} / 0.01
Epochs per round	1
Global / micro batch size	8 / 1
Precision / DeepSpeed	bfloat16 / ZeRO Stage 2
LoRA rank / scaling / dropout	64 / 128 / 0.1
LoRA target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj

for $0 < \lambda < 2/L_F$. This concerns a fixed local objective; it does not cover stochastic optimizer steps or establish convergence across aggregation and curriculum changes.

C Training Configuration

Table 8 lists the reported settings. Mixed precision, gradient checkpointing, and FlashAttention-2 support memory-efficient training.

D Supplementary Experimental Results

Client-level round trajectories. Tables 9 and 10 provide the client-level values underlying the aggregate trajectories. MATH improves in six of seven subjects from Round 2 to Round 3; Intermediate Algebra changes from 53.50% to 53.20%. All six CodeXGLUE clients improve over the three rounds.

Table 9: MATH accuracy (%) by communication round. Avg. is the subject macro average.

Task	R1	R2	R3
Algebra	84.30	89.60	91.20
Counting	51.30	65.20	69.60
Geometry	50.70	59.10	60.80
Inter. Alg.	51.10	53.50	53.20
Num. Theory	65.60	72.40	74.40
Prealgebra	74.10	83.10	86.20
Precalc.	52.20	55.30	56.40
Avg.	61.33	68.31	70.26

Table 10: CodeXGLUE BLEU-4 by communication round. Avg. is the language macro average.

Language	R1	R2	R3
Go	8.52	9.13	12.06
Java	9.34	9.98	11.06
JS	6.32	6.49	8.53
PHP	13.85	14.23	14.68
Py.	59.47	79.80	90.81
Ruby	6.33	6.73	8.27
Avg.	17.31	21.06	24.24

Client-Specific Selection Patterns. Figure 7 provides a client-level view of the selection process. Selected and unselected distributions differ across clients and rounds, showing that a single global threshold would not reproduce the resulting curricula. Later rounds admit medium and hard pairs without uniformly favoring the nominally hardest group. Since no external correctness labels are used in these plots, they characterize selection behavior rather than data quality.

CodeXGLUE curriculum visualization. Figure 6 complements the MATH visualization with programming-language clients. The selected distributions vary across clients and rounds, illustrating personalized selection within the shared availability schedule.

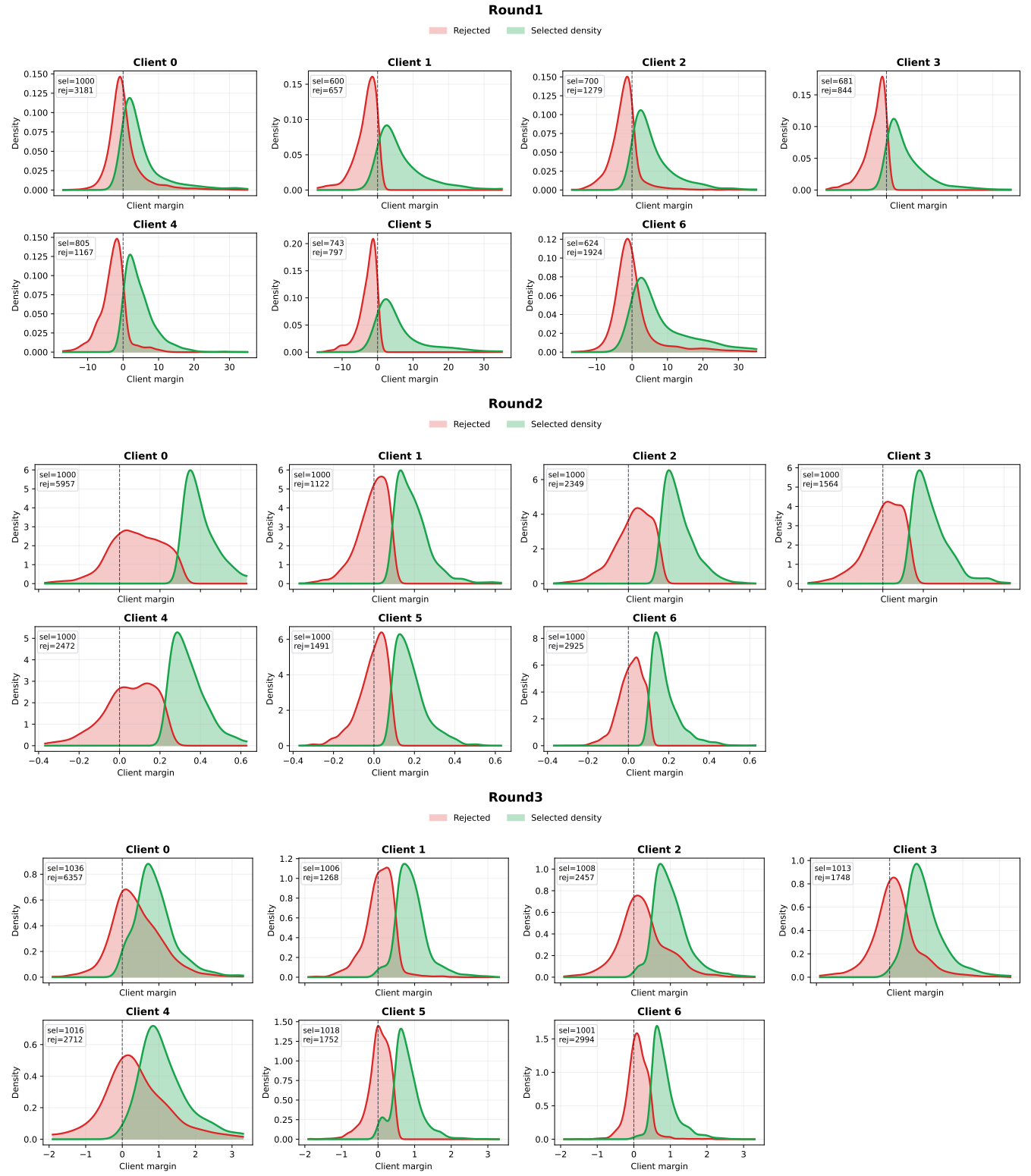


Figure 7: Selected (green) and unselected (red) candidate-margin distributions across clients and communication rounds. Their client-dependent separation reflects personalized greedy ranking.